RAJALAKSHMI ENGINEERING COLLEGE

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Started on	Thursday, 6 August 2026, 10:48 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 4:52 PM
Time taken	19 days 6 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{  int n;
    scanf("%d", &n);
    int i = 1;
    int s = 1;
    int counter = 2;
    while (s <= n)
    {
        counter++;      // i++
        i++;
        counter += 2;   // s = s + i  (addition + assignment)
        s += i;
    }
   counter++;
    printf("%d", counter);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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